Study-aware QSPR of human skin permeability: separating molecular signal from source heterogeneity and calibrating unseen sources

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1 Abstract

Human skin permeability measurements are commonly pooled across studies even though permeability depends not only on molecular properties but also on study-specific experimental conditions. We tested whether this heterogeneity can be modeled as a reusable source-associated component rather than treated solely as noise. Three public human-skin permeability resources were used sequentially rather than pooled: the Cheruvu et al. 2022 dataset for development, HuSkinDB for independent replication and held-out-source calibration, and SkinPiX for cross-dataset transfer. Measurement-level mixed-effects models separated molecular, residual compound-specific, literature-source, and residual variation. The molecular component was deliberately simple and prespecified, using molecular weight and logP alone (S1) or a six-descriptor physicochemical specification (S2). In Cheruvu et al., source-associated variation accounted for 22.8% of modeled random/residual variance; in HuSkinDB it accounted for 44.4%. Source-blind hierarchical prediction improved leave-one-compound-out RMSE by only 5.4–7.0% in Cheruvu and did not improve HuSkinDB, whereas known-source prediction improved RMSE by 16.7–18.7% in Cheruvu and 17.0% in HuSkinDB. When entire HuSkin sources were held out, offsets estimated from shared anchor compounds improved prediction of compounds absent from model fitting; among nine matched sources, mean source-level RMSE reduction increased from 31.1% with one anchor to 45.3% with five anchors. A falsification control produced the opposite result: offsets borrowed from other sources worsened prediction. Finally, a model trained only on HuSkinDB transferred to four independent SkinPiX sources. One-anchor calibration reduced mean source-level RMSE by 17.6% across four independent SkinPiX sources containing 54 HuSkin-unseen compounds, all within the HuSkin-defined chemical applicability domain. These findings support a study-aware QSPR framework in which literature-source heterogeneity is modeled explicitly and can be estimated for a new experimental source using a small number of shared compounds.

Keywords: skin permeability; QSPR; mixed-effects model; hierarchical model; source calibration; external transfer; applicability domain; HuSkinDB; SkinPiX

2 Introduction

The skin permeability coefficient, K_p, is a central parameter in transdermal drug delivery, dermal toxicology, cosmetics, and exposure assessment. Because direct measurement in human skin is expensive, tissue-limited, and experimentally demanding, quantitative structure–property relationship (QSPR) models have long been used to estimate permeability from molecular structure. The classical Potts–Guy model relates permeability primarily to lipophilicity and molecular size (Potts and Guy 1992), while later studies have explored broader physicochemical descriptor sets and more flexible statistical or machine-learning methods (Tsakovska et al. 2017; Zeng et al. 2021).

A major complication is that the published literature does not represent a single experimental system. Human-skin permeability values are generated across laboratories, diffusion-cell configurations, skin sources and anatomical sites, tissue-preparation procedures, donor and receptor media, temperatures, analytical methods, and reporting conventions. Interstudy variation has therefore been recognized for decades. Fujiwara et al. modeled a latent common permeability underlying study-specific measurements (Fujiwara et al. 2003). Abraham and Martins standardized literature values for temperature and ionization before fitting linear free-energy relationships (Abraham and Martins 2004). Samaras et al. showed that experimental and formulation conditions materially influence in-vitro human skin permeation (Samaras et al. 2012). More recently, HuSkinDB and SkinPiX have made measurement-level permeability records and protocol metadata available in structured public resources (Stepanov et al. 2020; Chedik et al. 2024).

Recent predictive studies further narrow the remaining methodological gap. QSPR models have been developed from curated HuSkinDB records (Waters and Quah 2022), and newer work has combined HuSkinDB and SkinPiX into larger curated resources for predictive modeling (Asgarkhanova et al. 2026). Oh and Ahn explicitly examined disagreement among repeated permeability measurements, protocol variables, record-level versus compound-level modeling, and mixed-effects formulations (Oh and Ahn 2026). Consequently, novelty can no longer rest on the general observations that repeated measurements disagree, that experimental conditions matter, or that mixed-effects models can be fitted to permeability data.

The more operational question is whether source-associated heterogeneity can be **estimated for a previously unseen experimental source and then used to improve predictions for new compounds measured by that source**. If measurements can be separated into a transferable molecular component and a source-associated component shared across compounds, then compounds measured both in a reference corpus and on a new platform could serve as calibration anchors. Under this interpretation, study heterogeneity becomes not only a source of uncertainty but also a parameter that can be estimated when deploying a model to a new laboratory, apparatus, or protocol.

We therefore considered the measurement-level model

$$y_{ij} = f(X_i) + \beta^\top Z_{ij} + b_i + u_j + \epsilon_{ij},$$

where y_{ij} is log permeability for compound i measured in source j, $f(X_i)$ is the systematic molecular component, Z_{ij} contains measured experimental covariates, b_i is residual compound-specific variation not explained by $f(X_i)$, u_j is a source-associated random intercept, and ϵ_{ij} is measurement-level residual error. For a previously measured compound, the total chemical contribution is $f(X_i) + b_i$; for a genuinely unseen compound, only $f(X_i)$ is available in expectation.

The analysis was designed as development, independent replication, and cross-dataset transfer. The Cheruvu et al. dataset (Cheruvu et al. 2022) was used to establish the variance structure and compare conventional compound aggregation with measurement-level hierarchical prediction. HuSkinDB was then used to replicate the hierarchy and to simulate deployment by holding out entire literature sources. Finally, SkinPiX was kept separate as an external transfer set: the molecular model was trained on HuSkinDB only, shared HuSkin–SkinPiX compounds were used as anchors, and SkinPiX-only compounds were reserved for evaluation. The central hypotheses were that (i) permeability measurements contain a reproducible source-associated component not captured by molecular descriptors alone, and (ii) this component can be estimated from a small number of shared compounds to improve prediction at an unseen source.

3 Materials and Methods

3.1 Study design and data sources

Three public human-skin permeability datasets were analyzed sequentially (Table 1). The Cheruvu et al. 2022 compilation was used for model development (Cheruvu et al. 2022). After removal of three rows without compound identity, it contained 476 measurements from 148 compounds and 69 literature references.

HuSkinDB was used for independent replication and internal source-calibration experiments (Stepanov et al. 2020); the processed dataset contained 546 measurements from 251 compounds and 95 references. SkinPiX was used only for cross-dataset transfer (Chedik et al. 2024). Of 202 original records, two non-equality permeability records and one missing or non-positive processed K_p record were excluded, leaving 199 measurements from 108 compounds and 36 references.

Table 1: Dataset structure and role in the analysis.

Dataset	Measurements	Compounds	Sources	Repeated compounds	Multi-source compounds	Role
Cheruvu	476	148	69	81	46	Development
HuSkinDB	546	251	95	89	61	Independent replication and held-out-source calibration
SkinPiX	199	108	36	65	14	Cross-dataset transfer validation

The datasets were deliberately **not pooled before validation**. Their separation allowed the analysis to distinguish development from replication and to evaluate transfer to an independently curated database. This design also reduced the risk that apparently external performance would be driven by duplicate literature sources present in more than one database.

3.2 Harmonization and compound/source identities

In the Cheruvu dataset, canonical compound identity was based primarily on normalized compound name rather than CAS number because the source table contained at least one CAS collision between chemically distinct entries. Reference identity was taken from the literature-source field. Permeability values were analyzed on the supplied log scale. Experimental temperature values reported in Kelvin were converted to degrees Celsius.

HuSkinDB compound identity was defined using SMILES where available, with normalized compound names used as fallback. Literature sources were identified preferentially by DOI and otherwise by the database reference field. HuSkinDB reports log K_p in cm/s; values were converted to $\log_{10}$(cm/h) by adding $\log_{10}(3600)$ so that the endpoint was compatible with the other analyses.

For SkinPiX, compound identity was defined from SMILES with normalized names as fallback. Source identity was based on publication identifier and DOI. Exact equality measurements were retained; censored inequality records were excluded rather than treated as exact K_p values. Cross-database HuSkin–SkinPiX compound overlap was identified first by exact SMILES and then by unambiguous normalized-name matching. All resulting overlap assignments were manually audited. No DOI overlap was identified between the HuSkin and SkinPiX sources used for transfer.

Because HuSkinDB and SkinPiX did not provide identical molecular-descriptor definitions, molecular weight and Crippen logP were recomputed from SMILES using RDKit for **both** databases (Landrum et al. 2026). All HuSkin and SkinPiX structures parsed successfully. Among the 24 matched compounds, the median absolute difference in RDKit molecular weight and logP between the database representations was zero, supporting descriptor compatibility across the transfer analysis.

3.3 Hierarchical decomposition in the development dataset

The initial Cheruvu hierarchy was examined using nested linear mixed-effects models fitted with `lme4` (Bates et al. 2015):

$$
\begin{aligned}
M_0 &: y_{ij} = \alpha + b_i + \epsilon_{ij}, \\
M_1 &: y_{ij} = \alpha + b_i + u_j + \epsilon_{ij}, \\
M_2 &: y_{ij} = \alpha + \beta_T T_{ij} + b_i + u_j + \epsilon_{ij}, \\
M_3 &: y_{ij} = \alpha + \beta_T T_{ij} + \beta_H H_{ij} + b_i + u_j + \epsilon_{ij},
\end{aligned}
$$

where T_{ij} is donor/skin-surface temperature and H_{ij} is skin thickness. Compound and reference were modeled as crossed random intercepts. Maximum-likelihood fits were used to compare fixed-effect specifications; restricted maximum likelihood (REML) fits were used for final parameter and variance-component estimation. For presentation, the compound, source, and residual variance components were expressed as proportions of their sum, $\sigma_b^2 + \sigma_u^2 + \sigma_\epsilon^2$. These percentages therefore describe the partitioning of modeled random/residual variance and do not include variation attributable to fixed effects. Because testing a variance component against zero places the null hypothesis on the parameter boundary, the addition of the reference random effect was interpreted primarily using the large change in information criteria and the estimated variance component rather than relying on a naive chi-square test. Temperature and skin-thickness additions were compared using ordinary likelihood-ratio tests.

3.4 Molecular QSPR specifications

The molecular component $f(X)$ was prespecified rather than selected by stepwise search. Two specifications were evaluated in Cheruvu:

S1: size–lipophilicity model

$$
f_{S1}(X_i) = \beta_0 + \beta_1 MW_i + \beta_2 \log K_{ow,i}.
$$

S2: extended physicochemical model

$$
f_{S2}(X_i) = \beta_0 + \beta_1 MW_i + \beta_2 \log K_{ow,i} + \beta_3 MP_i + \beta_4 \log S_{aq,i} + \beta_5 HBD_i + \beta_6 HBA_i.
$$

Descriptors were standardized using compound-level training-set means and standard deviations. The molecular terms were fitted jointly with the selected hierarchical structure rather than estimated from pre-aggregated permeability values. The explanatory contribution of $f(X)$ was summarized as the proportional reduction in compound random-effect variance relative to the hierarchy-only model.

For conventional QSPR comparison, measurements were collapsed to a mean log permeability per compound and molecular descriptors were summarized by their compound median. This provided a conventional compound-level benchmark while keeping the molecular information matched as closely as possible to the hierarchical models.

HuSkinDB replication used S1 only because molecular weight and logP were complete for all records, whereas donor temperature was missing in 155 of 546 measurements and the broader Cheruvu descriptor panel was not available in directly comparable form. Temperature was therefore evaluated as a secondary complete-case analysis in HuSkinDB rather than as a required term in the primary HuSkin QSPR.

Accordingly, variance decompositions from Cheruvu et al. and HuSkinDB were used to assess replication of a substantial source-associated component, not to compare the absolute magnitude of variance components directly between datasets. The Cheruvu decomposition adjusts for temperature as a fixed effect, whereas the primary HuSkinDB decomposition does not.

3.5 Leave-one-compound-out validation and prediction estimands

Prediction was evaluated using leave-one-compound-out (LOCO) validation. In each fold, all measurements of one compound were removed before model fitting and descriptor standardization. Three estimands were distinguished explicitly:

1. **Conventional aggregate QSPR:** prediction from the compound-level regression.
2. **Hierarchical, source-blind:** prediction from the hierarchical molecular fixed effects with all random effects set to zero for the held-out compound.
3. **Hierarchical, known-source:** prediction using the molecular fixed effects plus a source random effect when the held-out measurement belonged to a source represented in the training fold. New source levels received zero effect.

This distinction was critical because a known-source hierarchical predictor has access to information unavailable to the conventional aggregate model. The source-blind prediction therefore served as a direct test of whether hierarchical estimation itself produced a superior transferable molecular function $f(X)$, whereas the known-source prediction quantified the additional value of source context.

Performance was summarized using RMSE, MAE, predictive R^2, and compound-balanced RMSE. Sensitivity analyses in Cheruvu examined exclusion of observations with absolute standardized residuals greater than three, descriptor aggregation by mean rather than median, restriction to multi-reference compounds, and an expanded nonlinear molecular specification. These analyses were secondary and did not alter the prespecified primary specifications.

3.6 Chemical applicability domain

Chemical applicability was assessed using descriptors only, separately from experimental-source applicability. For every LOCO fold, descriptor means, standard deviations, covariance matrices, and univariate ranges were estimated **from the training compounds only**. A held-out compound was considered in-domain when (i) its squared Mahalanobis distance was below the 95th percentile of the appropriate chi-square distribution and (ii) every descriptor remained within the corresponding univariate training range. S1 therefore used a two-dimensional domain and S2 a six-dimensional domain.

For the SkinPiX transfer analysis, applicability was defined relative to the complete HuSkin training descriptor space because HuSkinDB was the deployed reference model and SkinPiX was external. No SkinPiX information was used to define the chemical domain.

3.7 Held-out-source calibration in HuSkinDB

To simulate deployment to an unseen experimental source, each HuSkin reference was held out in turn. The model was fitted to all other references. Compounds present both in the held-out source and in the training data were considered candidate **anchor compounds**. Compounds present in the held-out source but absent from all model-fitting data were reserved as genuinely unseen test compounds.

For an anchor compound i measured in held-out source L, the residual relative to the fitted reference model was

$$r_{iL} = y_{iL} - \hat{\mu}_i,$$

where $\hat{\mu}_i$ included the molecular fixed effect and the known compound-specific random effect estimated from the training sources. Repeated anchor measurements were first averaged within compound. For a panel of k anchors, the raw source offset was the mean anchor residual,

$$\bar{r}_L = \frac{1}{k} \sum_{i=1}^{k} r_{iL}.$$

The estimated source effect was shrunk toward zero using the random-effect and residual variances from the training model,

$$\hat{u}_L = \frac{k\hat{\sigma}_u^2}{\hat{\sigma}_\epsilon^2 + k\hat{\sigma}_u^2} \bar{r}_L.$$

Predictions for unseen compounds in the held-out source were then shifted by $\hat{u}_L$. Anchor-panel sizes from one to five compounds were evaluated where available. For each panel size, 100 random anchor selections were drawn with a fixed random seed.

Prediction error was first calculated separately within each held-out source. For source s and anchor-selection replicate r, RMSE was calculated across the unseen test measurements from that source. For each anchor-panel size, these source-specific RMSE values were averaged across the 100 anchor selections and then across eligible sources, giving equal weight to each experimental source regardless of the number of test measurements it contributed. We refer to this quantity as the mean source-level RMSE. Percentage improvement from calibration was calculated as

$$100 \left(1 - \frac{\overline{\text{RMSE}}_{\text{calibrated}}}{\overline{\text{RMSE}}_{\text{uncalibrated}}} \right).$$

Results were summarized both across all sources eligible for a given panel size and in matched source cohorts supporting the same panel sizes. Because the set of eligible sources changes as the required number of anchors increases, inference about anchor number was based primarily on matched cohorts. The nine-source cohort supporting all panel sizes from one to five anchors was used for the primary HuSkin anchor-number comparison.

3.8 Falsification by wrong-source calibration

A negative control tested whether improved predictions could arise merely from adding an arbitrary offset. For each held-out HuSkin source and anchor-panel size, the correctly estimated source offset was compared with an offset generated from a **different** held-out source with the same anchor-panel size. The novel-compound test set itself was unchanged. If calibration reflects a source-specific latent component, the matched source offset should improve prediction whereas a borrowed offset should not. Source-matched and different-source calibration were evaluated using the same mean source-level RMSE and percentage-improvement definition described above, with comparisons made within the same eligible source set at each anchor-panel size.

3.9 Cross-dataset transfer from HuSkinDB to SkinPiX

The SkinPiX analysis was designed as a stricter cross-dataset transfer test. A hierarchical S1 model was fitted to HuSkinDB using RDKit molecular weight and logP. SkinPiX was not used for model fitting. Within each SkinPiX source, compounds also present in HuSkinDB were candidate anchors, while SkinPiX-only compounds were the test set. A source was eligible when it contained at least one anchor, at least one SkinPiX-only compound, and no DOI overlap with HuSkinDB.

Cross-dataset calibration performance was summarized using the same source-balanced procedure as in the HuSkin held-out-source experiment. RMSE was calculated separately for the SkinPiX-only test measurements within each source and each anchor-selection replicate, averaged across the 100 anchor selections, and then averaged across sources. Thus, each SkinPiX source contributed equally to the reported mean source-level RMSE regardless of its number of test measurements.

The same empirical-Bayes calibration procedure used in the HuSkin internal experiment was then applied to each eligible SkinPiX source. One- through five-anchor panels were evaluated where possible with 100 random anchor selections. Because only four SkinPiX sources simultaneously contained anchors and genuinely novel compounds, the one-anchor result across all four sources was treated as the primary external-transfer result. A matched three-source subset supporting one through three anchors was used only as a secondary anchor-number sensitivity analysis. Four- and five-anchor results involved progressively fewer sources and were treated descriptively rather than as evidence of a general dose-response.

For compound-level visualization of the one-anchor experiment, calibrated predictions were averaged across the 100 random one-anchor selections for each measurement. This visualization therefore represents expected performance over random selection of one available anchor rather than a single privileged anchor choice.

3.10 Software and reproducibility

All statistical analyses were performed in R. Linear mixed-effects models were fitted with `lme4` (Bates et al. 2015). RDKit was used in a separate Python environment to calculate molecular weight and Crippen logP from SMILES (Landrum et al. 2026). Analysis scripts were organized sequentially so that data preparation, model fitting, validation, falsification, and manuscript output were separated and reproducible. No results were recalculated in the manuscript-generation scripts.

4 Results

4.1 Source-associated heterogeneity replicates across datasets

Adding a literature-source random intercept to the Cheruvu hierarchy reduced AIC from 1033.64 to 823.29. Adding temperature further reduced AIC to 812.32, whereas adding skin thickness increased AIC slightly to 813.66. The temperature coefficient was 0.0425 $\log_{10}$ units per °C ($p =< 0.001$), corresponding to approximately a 10.3% multiplicative increase in K_p per degree Celsius. Skin thickness was not supported ($p = 0.42$). The M2 model was therefore used as the primary Cheruvu hierarchy.

In the REML variance decomposition, the random/residual variance in Cheruvu et al. was partitioned into compound-associated (66.4%), source-associated (22.8%), and residual (10.8%) components (Table 2; Figure 1). HuSkinDB independently reproduced substantial source-associated variation, with the source component accounting for 44.4% of its modeled random/residual variance. Because the Cheruvu model included temperature as a fixed effect whereas the primary HuSkinDB model did not, these percentages should not be interpreted as a direct quantitative comparison between datasets. Rather, the replicated result is that a substantial source-associated component was present in both datasets.

Table 2: Partitioning of modeled random/residual variance. Percentages are normalized over the compound, source, and residual variance components; fixed-effect variation is not included in the denominator. The Cheruvu and HuSkinDB decompositions use different fixed-effect specifications and should therefore be interpreted as independent evidence of substantial source-associated variation rather than compared directly in magnitude.

Dataset	Variance component	Estimate (%)	95% CI
Cheruvu	Compound	66.4	[56.9, 74.7]
Cheruvu	Source	22.8	[14.4, 31.9]
Cheruvu	Residual	10.8	[8.3, 14.0]
HuSkinDB	Compound	23.5	[15.2, 32.9]
HuSkinDB	Source	44.4	[33.5, 54.5]

Dataset	Variance component	Estimate (%)	95% CI
HuSkinDB	Residual	32.1	[24.7, 40.8]

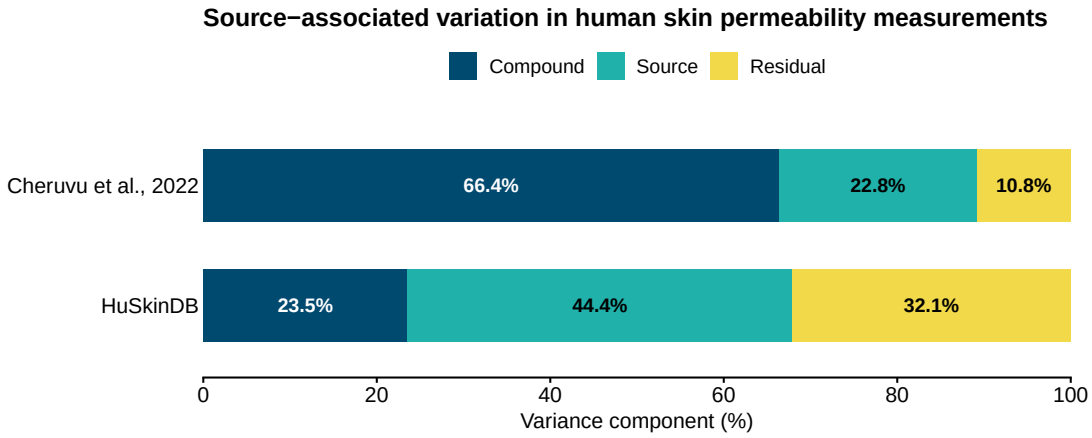


Figure 1: Variance partitioning in Cheruvu et al. 2022 and HuSkinDB. Bars show the relative contributions of compound, literature source, and residual components to modeled random/residual variance. The Cheruvu decomposition adjusts for temperature, whereas the primary HuSkinDB decomposition does not because temperature was incomplete; the figure therefore demonstrates replication of substantial source-associated variation rather than a direct comparison of component magnitudes between datasets.

4.2 Molecular descriptors explain substantial compound variation, but source context drives most of the predictive gain

In Cheruvu, S1 (molecular weight plus logKow) explained 60.3% of the compound random-effect variance. The broader S2 specification explained 61.4%, only 1.1 percentage points more. Thus, most of the explainable compound signal captured by these prespecified models was already present in the simple size–lipophilicity specification. In HuSkinDB, molecular weight plus logP explained 37.1% of between-compound variance.

LOCO validation separated improvement in the molecular component from improvement due to source context. For Cheruvu S1, conventional aggregate QSPR produced RMSE 0.866. Source-blind hierarchical prediction reduced RMSE to 0.819, a 5.4% improvement. Allowing known source information reduced RMSE further to 0.721, a 16.7% improvement. S2 showed the same pattern: source-blind hierarchical prediction improved RMSE by 7.0%, whereas known-source prediction improved it by 18.7%.

HuSkinDB provided the stricter result. Conventional S1 RMSE was 1.333, while source-blind hierarchical RMSE was 1.355, corresponding to a -1.7% change. Thus, hierarchical estimation did **not** improve the source-independent molecular predictor in HuSkinDB. In contrast, known-source hierarchical prediction reduced RMSE to 1.106, a 17.0% improvement. The large prediction gain therefore arose primarily from correctly representing source context rather than from fitting a universally superior source-independent molecular QSPR.

Table 3: Leave-one-compound-out prediction under conventional, source-blind hierarchical, and known-source hierarchical settings. Confidence intervals were obtained by paired compound-level bootstrap resampling of held-out compounds.

Dataset	Specification	Prediction setting	RMSE (95% CI)	RMSE improvement vs conventional (%)
Cheruvu	S1	Conventional aggregate QSPR	0.866 [0.725, 1.017]	Reference
Cheruvu	S1	Hierarchical, source blind	0.819 [0.708, 0.945]	5.4 [-3.6, 10.9]
Cheruvu	S1	Hierarchical, known source	0.721 [0.601, 0.855]	16.7 [11.6, 21.8]
Cheruvu	S2	Conventional aggregate QSPR	0.850 [0.715, 0.977]	Reference
Cheruvu	S2	Hierarchical, source blind	0.790 [0.703, 0.883]	7.0 [-5.1, 14.0]
Cheruvu	S2	Hierarchical, known source	0.691 [0.602, 0.788]	18.7 [11.5, 23.3]
HuSkinDB	S1	Conventional aggregate QSPR	1.333 [1.198, 1.474]	Reference
HuSkinDB	S1	Hierarchical, source blind	1.355 [1.201, 1.537]	-1.7 [-5.1, 1.0]
HuSkinDB	S1	Hierarchical, known source	1.106 [0.959, 1.305]	17.0 [10.0, 23.8]

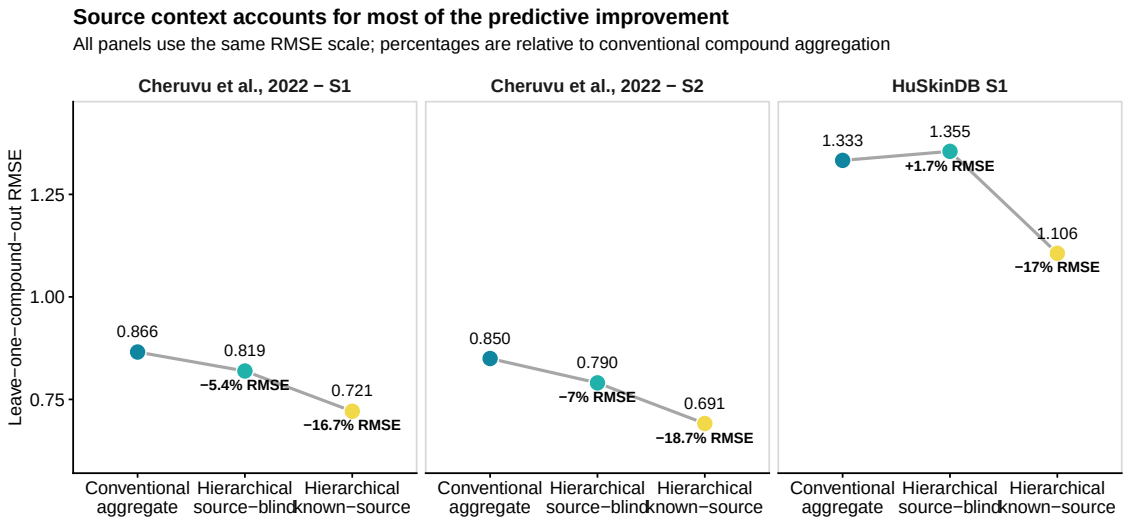


Figure 2: Leave-one-compound-out RMSE under conventional aggregate QSPR, source-blind hierarchical prediction, and hierarchical prediction with known source context. Common y-axis limits show that source-blind gains were modest in Cheruvu et al. and absent in HuSkinDB, whereas source context produced the larger and reproducible reduction in prediction error.

4.3 Shared compounds calibrate completely held-out HuSkin sources

Holding out entire HuSkin references tested whether source information could be recovered rather than assumed known. Calibration used only compounds shared between the held-out source and the training corpus; evaluation was restricted to compounds from that source that were absent from all model-fitting data.

Calibration improved mean source-level RMSE at every tested anchor-panel size among eligible sources. Because eligibility changed with panel size, the primary anchor-number comparison used the same nine references that supported one through five anchors. Within this matched cohort, mean source-level RMSE decreased by 31.1% with one anchor and by 45.3% with five anchors. Because the same nine sources contributed at every panel size, this monotonic trend is not attributable to changing source composition.

The cohort-level trend did not imply universal source-level success. At five anchors, 8 of nine matched sources improved, while HUR_48 deteriorated substantially. The median source-specific RMSE improvement at five anchors was 40.0%, whereas the reduction in mean source-level RMSE was 45.3%. These quantities differ because the latter is calculated from the ratio of the mean calibrated and uncalibrated source-level RMSEs, rather than as the arithmetic mean or median of source-specific percentage improvements.

Table 4: Held-out HuSkin source calibration in the matched nine-source cohort.

Anchors	Matched sources	RMSE uncalibrated	RMSE calibrated	RMSE improvement (%)
1	9	1.135	0.782	31.1
2	9	1.135	0.705	37.9
3	9	1.135	0.675	40.5
4	9	1.135	0.641	43.6
5	9	1.135	0.621	45.3

4.4 Wrong-source calibration falsifies a generic-offset explanation

The wrong-source negative control produced the opposite behavior from source-matched calibration. With one through five anchors, source-matched calibration reduced mean source-level RMSE by 9.3%, 15.8%, 21.0%, 36.2%, and 45.3%, respectively. Applying an offset estimated from a different source instead changed mean source-level RMSE by -9.0%, -10.9%, -15.8%, -16.1%, and -10.3%, respectively, where negative improvement denotes worse prediction. Because the eligible source set changed with anchor-panel size, these values primarily compare source-matched with different-source calibration within each panel size; the across-panel trend should not itself be interpreted as an anchor-number effect.

This result makes several artifact explanations unlikely. The improvement was not generated merely by adding another fitted parameter to the prediction, because an equally structured offset from the wrong source was harmful. Nor is the effect readily explained by regression to the mean: the calibration signal had to be associated with the same experimental source as the target measurements. The source effect therefore contained information that was shared across compounds within a source and recoverable from anchor compounds.

Table 5: Falsification control comparing source-matched calibration with an offset borrowed from a different held-out HuSkin source.

Anchors	Eligible sources	Matched-source RMSE improvement (%)	Different-source RMSE improvement (%)
1	31	9.3	-9.0
2	22	15.8	-10.9
3	19	21.0	-15.8
4	12	36.2	-16.1

Anchors	Eligible sources	Matched-source RMSE improvement (%)	Different-source RMSE improvement (%)
5	9	45.3	-10.3

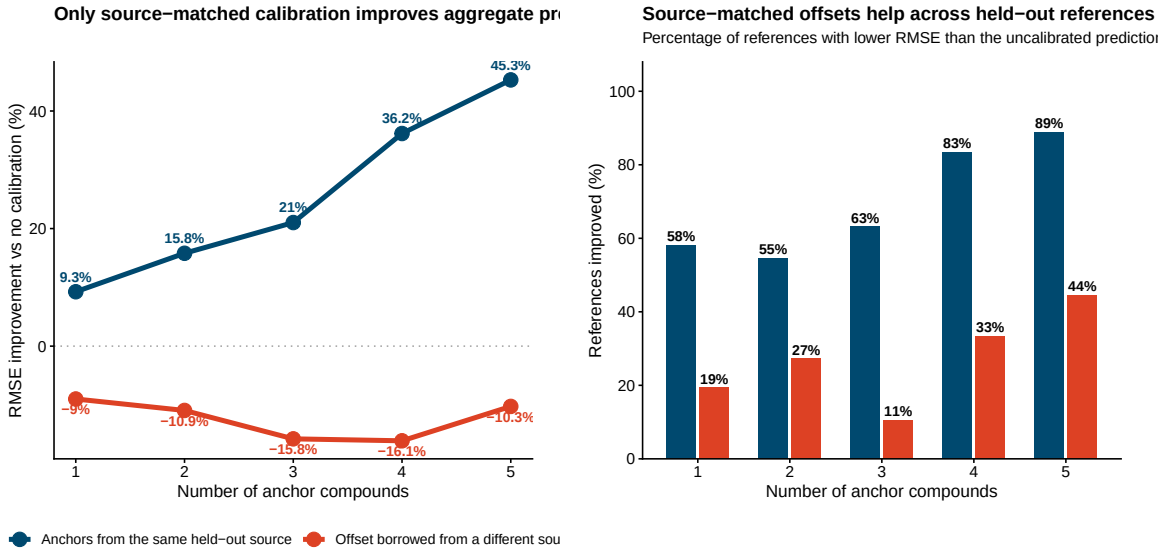


Figure 3: Falsification of a generic-offset explanation. Panel A shows mean source-level RMSE improvement when calibration is estimated from anchors measured in the same held-out source versus an offset borrowed from a different source. Panel B shows the proportion of held-out references improved by each procedure. Source-matched calibration helps; mismatched source information reverses the benefit.

For the compound-level visualization of the one-anchor experiment, calibrated predictions were first averaged across the 100 random one-anchor selections for each test measurement. Absolute errors were then calculated from these mean predictions and averaged within compound where necessary. This compound-level summary was used only for visualization and for counting compounds with lower absolute error after calibration; it is distinct from the mean source-level RMSE used for the primary calibration-performance estimate.

4.5 HuSkin-trained calibration transfers to independent SkinPiX sources

The cross-dataset analysis tested the same mechanism under a stricter separation between model development and deployment. The molecular model was fitted to 546 HuSkin measurements from 251 compounds. SkinPiX contained 24 compounds overlapping HuSkinDB and 84 SkinPiX-only compounds. Fifteen SkinPiX sources contained at least one HuSkin-overlapping compound, but only four contained both an available anchor and at least one SkinPiX-only compound suitable for genuinely external evaluation. Across these four sources, the test set comprised 54 unique compounds not present in HuSkinDB.

With one randomly selected anchor compound per source, calibration reduced mean source-level RMSE by 17.6% across the four eligible SkinPiX sources. Because RMSE was calculated within source and then averaged across sources, each source contributed equally to this estimate regardless of its number of test measurements. At the source level, 3 of four sources improved. At the compound level, 44 of 54 unseen SkinPiX compounds had lower mean absolute error after calibration when predictions were averaged across the 100 one-anchor selections. The source-level RMSE and compound-level absolute-error summaries therefore describe complementary but distinct aspects of transfer performance.

Three of the four eligible SkinPiX sources supported at least three anchors. Restricting the analysis to these same three sources, mean source-level RMSE improvement increased from 18.6% with one anchor to 22.1% with two and 23.9% with three. This small matched subset provides supportive rather than definitive evidence for an anchor-number effect across databases. Four- and five-anchor SkinPiX estimates were not emphasized because they involved only two and one sources, respectively.

All 54 external SkinPiX test compounds were inside the HuSkin-defined chemical applicability domain. The transfer result should therefore be interpreted primarily as **experimental-source transfer within HuSkin chemical space**, not as evidence for reliable chemical extrapolation beyond that space.

Table 6: Cross-dataset calibration of SkinPiX sources using a model trained only on HuSkinDB.

Cohort	Anchors	Sources	RMSE uncalibrated	RMSE calibrated	RMSE improvement (%)
All eligible sources	1	4	0.810	0.667	17.6
Matched three-source cohort	1	3	0.949	0.772	18.6
Matched three-source cohort	2	3	0.949	0.740	22.1
Matched three-source cohort	3	3	0.949	0.722	23.9

4.6 Applicability-domain diagnostics

Applicability-domain analysis was used as a deployment qualification rather than as an additional model-selection criterion. In Cheruvu, 137 of 148 compounds were classified in-domain under S1 and 128 of 148 under S2. In HuSkinDB, 243 of 251 compounds were in-domain under S1. All 54 SkinPiX compounds used in the external transfer test were in-domain relative to the HuSkin training set. Domain-stratified errors and normalized fold-specific Mahalanobis distances are reported in Supplementary Table S7 and Supplementary Figure S1.

5 Discussion

5.1 Source-associated heterogeneity is a reproducible part of permeability data

The central result is not simply that a mixed-effects model fits the literature better. Rather, two independently curated datasets show that a substantial component of measurement variability is associated with literature source, and that this source-associated component carries predictive information across compounds measured within the same source. Source-associated variation represented 22.8% of modeled random/residual variance in Cheruvu et al. and 44.4% in HuSkinDB. Because the two decompositions were not based on identical fixed-effect specifications, their numerical magnitudes should not be compared directly. The important replicated finding is that source-associated variation constituted a substantial component in both independently curated datasets.

This result is consistent with decades of evidence that permeability depends on experimental context (Fujiwara et al. 2003; Abraham and Martins 2004; Samaras et al. 2012). The present analysis does not claim that heterogeneity is newly discovered. Its contribution is to show that, under a simple hierarchical model, part of that heterogeneity behaves like a **calibratable source-level quantity** that can be estimated from shared compounds and then transferred to compounds not used in fitting.

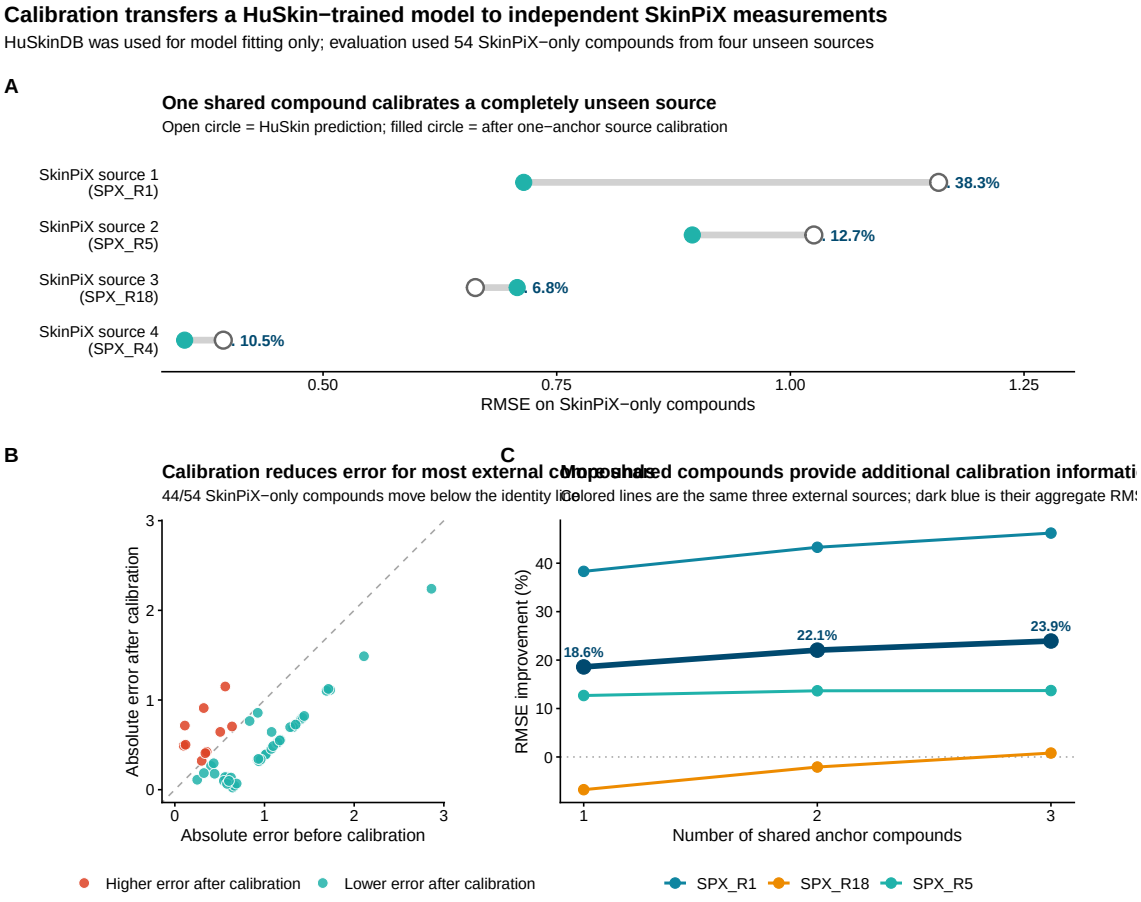


Figure 4: Cross-dataset transfer from HuSkinDB to SkinPiX. Panel A compares uncalibrated and one-anchor calibrated RMSE within each of the four eligible unseen SkinPiX sources. Panel B compares compound-level absolute prediction error before and after calibration for SkinPiX-only compounds. Panel C shows the matched three-source subset supporting one through three anchors; colored lines represent individual sources and the dark line represents reduction in mean source-level RMSE across those same sources.

5.2 Hierarchical fitting does not automatically create a better universal molecular QSPR

The source-blind falsification analysis narrows the interpretation of the LOCO results. In Cheruvu, retaining measurement-level hierarchy modestly improved the transferable molecular predictor by approximately 5–7%. In HuSkinDB, it did not: source-blind hierarchical prediction was slightly worse than conventional aggregation. The much larger 17–19% gains appeared only when source context was available.

This is scientifically useful because it argues against framing the method as a universally superior estimator of $f(X)$. The data instead support a two-part prediction problem. Molecular descriptors determine the expected permeability of an unseen compound, while source information shifts that expectation toward the scale of a particular experimental system. The value of the hierarchical formulation is therefore not that it always discovers a better molecular equation, but that it separates molecular prediction from source calibration.

The modest difference between Cheruvu S1 and S2 reinforces this point. Expanding from molecular weight and logKow to six descriptors increased explained compound variance by only about one percentage point. A more complex descriptor search might improve point prediction, but it would obscure the methodological question addressed here and increase the risk that performance differences were driven by feature selection rather than by the treatment of measurement hierarchy.

5.3 Anchor compounds convert an unseen source from unknown to calibratable

The held-out-source experiment is closest to the intended deployment setting. A new source initially has no estimated random effect, so prediction defaults to the reference scale. Once shared compounds are measured, their residuals relative to the reference model provide information about the new source's systematic displacement. Empirical-Bayes shrinkage prevents a small anchor panel from being interpreted as a perfectly known offset.

The matched nine-source HuSkin analysis showed a monotonic aggregate improvement as anchors increased from one to five, with diminishing returns after the first few compounds. This is practically encouraging because deployment may not require a large calibration panel. At the same time, HUR_48 demonstrates that a single source-wide intercept is not universally adequate. Possible explanations include compound-by-protocol interactions, condition changes within a publication, incorrect metadata grouping, formulation effects, or other deviations from the common-offset assumption. That failure case should be treated as part of the method's operating characteristics rather than removed post hoc.

Most importantly, the wrong-source control strongly supports the proposed mechanism. If calibration improvement arose from arbitrary recentering, flexible model fitting, or regression toward an overall mean, an offset drawn from another source should frequently provide similar benefit. Instead, source-matched offsets improved prediction while borrowed offsets worsened it. The information recovered from anchors is therefore specific to the experimental source being calibrated.

5.4 Cross-dataset transfer provides the strongest deployment-oriented evidence

SkinPiX was intentionally withheld from model development. HuSkinDB provided the molecular model, SkinPiX-overlapping compounds supplied anchors, and SkinPiX-only compounds supplied the test set. No DOI overlap was identified between the two source sets used for transfer, and RDKit descriptors were recalculated from structure in both databases to avoid descriptor-definition mismatch.

The resulting one-anchor improvement of 17.6% across four SkinPiX sources is therefore qualitatively different from an internal cross-validation gain. It shows that a source-calibration procedure developed and validated inside one database can improve predictions in another independently curated database. The result should nevertheless be interpreted with the correct level of confidence. Only four SkinPiX sources were eligible for true anchor-plus-novel-compound testing, and one of the four did not improve with one-anchor calibration. The external evidence is therefore promising and mechanistically consistent with the HuSkin experiment, but

not yet large enough to establish the frequency with which calibration will succeed in arbitrary laboratories or protocols.

The fact that all 54 external test compounds were inside the HuSkin chemical domain is both a strength and a limitation. It strengthens the interpretation that the experiment tests **source transfer** rather than confounding source transfer with severe chemical extrapolation. Conversely, it means the present study provides no evidence that source calibration rescues predictions for compounds outside the molecular domain of the reference model. Chemical applicability and experimental-source applicability should therefore remain separate dimensions of deployment confidence.

5.5 Relation to previous work

Previous approaches have addressed permeability heterogeneity through standardization, latent common permeability, explicit experimental covariates, filtering, aggregation, and more recent record-level or mixed-effects analyses (Fujiwara et al. 2003; Abraham and Martins 2004; Samaras et al. 2012; Oh and Ahn 2026). The present framework is complementary rather than contradictory. Instead of attempting to remove all source variation before modeling, it retains measurement-level structure and asks whether the residual source component can be used operationally.

This distinction also separates the current analysis from studies whose primary objective is to maximize predictive performance from merged databases (Asgarkhanova et al. 2026; Oh and Ahn 2026). Pooling HuSkinDB and SkinPiX at the outset could produce a larger training set, but it would eliminate the clean cross-dataset transfer experiment. By preserving SkinPiX as an external source collection, the analysis tests a deployment behavior that would otherwise be difficult to identify.

5.6 Limitations

Several limitations constrain interpretation. First, the random reference effect is **source-associated**, not a causal estimate of laboratory, apparatus, operator, skin donor, or protocol effects individually. A publication-level intercept can absorb all systematic differences correlated with that source. The present results therefore do not justify statements such as "44% of permeability variation is caused by apparatus differences."

Second, a common random intercept assumes that a source shifts compounds approximately in parallel. The HUR_48 failure shows that this assumption can be violated. Future models could examine random slopes, protocol interactions, or hierarchical decomposition of source into laboratory, apparatus, tissue preparation, and other metadata where sufficiently complete data exist. Such extensions should be justified by new data rather than added solely to improve retrospective fit.

Third, the external SkinPiX validation is limited by source overlap. Although 54 novel compounds were evaluated, they came from only four eligible SkinPiX sources. The number of experimental sources, rather than the number of compound measurements alone, is therefore the main limitation on the generality of the cross-dataset conclusion.

Fourth, the chemical component was intentionally simple. The objective was to isolate the effect of measurement hierarchy and source calibration while holding molecular information relatively fixed, not to establish the highest-performing skin-permeability predictor available. More expressive molecular models could be substituted for $f(X)$, but they would require separate validation and should not be assumed to preserve the same calibration behavior.

Finally, the analyses are retrospective and rely on literature sources. The strongest validation would be prospective: fit and freeze the reference model, measure several chemically diverse anchor compounds on a previously unseen experimental platform, estimate its platform offset, and then predict additional compounds before observing their permeability. Such a study would directly test whether literature-derived source calibration transfers to a real new apparatus rather than another historical database.

5.7 Implications for experimental deployment

The framework suggests a practical deployment strategy. A reference model provides a molecular prediction on a common permeability scale. A new laboratory or platform then measures a small panel of anchor compounds already represented in the reference corpus. Their deviations from the reference expectation estimate a platform-specific offset, after shrinkage for limited anchor information. Predictions for new compounds can then be reported together with two separate applicability statements: whether the compound is chemically inside the reference domain and whether the experimental platform has been calibrated.

This creates four qualitatively different deployment states: (i) chemically in-domain and platform-calibrated, (ii) chemically in-domain but platform-uncalibrated, (iii) chemically out-of-domain but platform-calibrated, and (iv) chemically out-of-domain and platform-uncalibrated. The present study provides the strongest evidence for the first state and does not imply that source calibration compensates for chemical extrapolation.

6 Conclusions

Human skin permeability measurements contain substantial and reproducible source-associated variation that is not captured by molecular descriptors alone. Retaining measurement-level hierarchy does not consistently produce a better source-independent molecular QSPR, but it allows source context to be represented explicitly. When a source is known, this context materially improves prediction; when a source is new, shared anchor compounds can estimate a useful source offset.

The calibration effect survived two important tests. It improved prediction of compounds absent from model fitting when entire HuSkin sources were held out, and it failed when the offset was deliberately borrowed from the wrong source. The same approach then transferred from HuSkinDB to independent SkinPiX measurements, improving one-anchor prediction across 54 HuSkin-unseen compounds from four eligible sources. These findings support a study-aware QSPR framework in which molecular prediction and experimental-source calibration are treated as distinct but complementary components of permeability prediction.

7 Data and code availability

The analysis uses publicly available datasets from Cheruvu et al. 2022, HuSkinDB, and SkinPiX (Cheruvu et al. 2022; Stepanov et al. 2020; Chedik et al. 2024). Analysis scripts and processed outputs are maintained in the project repository. The final repository release should include the exact raw-data provenance, software-environment information for RDKit descriptor generation, and all scripts required to regenerate manuscript tables and figures.

8 Author contributions

Chayada Piantham: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review and editing.

Additional author contributions should be added according to the final author list and CRediT roles.

9 Funding

Funding information to be added before submission.

10 Competing interests

The author(s) declare no competing interests unless otherwise stated at submission.

11 Supplementary material

11.1 Supplementary Table S1. Cheruvu hierarchical model comparison

Table 7: Cheruvu hierarchical model comparison.

x
/Users/chayada_p/Data/CUDENT/202604_skin_permeability_qspr_model/R/tables/Table_S1_cheruvu_hierarchical_mod

11.2 Supplementary Table S2. Molecular QSPR specifications

Table 8: QSPR model summary and reduction in compound variance.

x
/Users/chayada_p/Data/CUDENT/202604_skin_permeability_qspr_model/R/tables/Table_S2A_qspr_model_summary.cs

Table 9: QSPR coefficient estimates.

x
/Users/chayada_p/Data/CUDENT/202604_skin_permeability_qspr_model/R/tables/Table_S2B_qspr_coefficients.csv

11.3 Supplementary Table S3. Cheruvu validation, applicability, and sensitivity analyses

Table 10: Cheruvu leave-one-compound-out performance.

x
/Users/chayada_p/Data/CUDENT/202604_skin_permeability_qspr_model/R/tables/Table_S3A_cheruvu_loco.csv

Table 11: Cheruvu applicability-domain performance summary.

x
/Users/chayada_p/Data/CUDENT/202604_skin_permeability_qspr_model/R/tables/Table_S3B_cheruvu_applicability.csv

Table 12: Cheruvu sensitivity analyses.

x
/Users/chayada_p/Data/CUDENT/202604_skin_permeability_qspr_model/R/tables/Table_S3C_cheruvu_sensitivity.csv

11.4 Supplementary Table S4. HuSkinDB replication details

Table 13: HuSkinDB replication summary.

x
/Users/chayada_p/Data/CUDENT/202604_skin_permeability_qspr_model/R/tables/Table_S4_huskin_replication.csv

11.5 Supplementary Table S5. HuSkin calibration across all eligible sources

Table 14: HuSkin held-out-source calibration across all sources eligible at each anchor-panel size.

x
/Users/chayada_p/Data/CUDENT/202604_skin_permeability_qspr_model/R/tables/Table_S5_huskin_all_eligible_calibr

11.6 Supplementary Table S6. HuSkin matched-source calibration by reference

Table 15: Source-level HuSkin calibration results for the matched nine-source cohort.

x
/Users/chayada_p/Data/CUDENT/202604_skin_permeability_qspr_model/R/tables/Table_S6_huskin_reference_calibrat

11.7 Supplementary Table S7. Applicability-domain summary

Table 16: Chemical applicability-domain summary across development, replication, and transfer analyses.

x
/Users/chayada_p/Data/CUDENT/202604_skin_permeability_qspr_model/R/tables/Table_S7_applicability_domain.csv

11.8 Supplementary Table S8. SkinPiX overlap and descriptor audit

Table 17: SkinPiX transfer and provenance audit summary.

x
/Users/chayada_p/Data/CUDENT/202604_skin_permeability_qspr_model/R/tables/Table_S8A_skinpix_transfer_audit.c

Table 18: HuSkin–SkinPiX compound-overlap audit.

x
/Users/chayada_p/Data/CUDENT/202604_skin_permeability_qspr_model/R/tables/Table_S8B_skinpix_compound_over

11.9 Supplementary Table S9. SkinPiX calibration by source

Table 19: SkinPiX cross-dataset calibration results by source and anchor-panel size.

x
/Users/chayada_p/Data/CUDENT/202604_skin_permeability_qspr_model/R/tables/Table_S9_skinpix_reference_calibrat

11.10 Supplementary Table S10. Wrong-source falsification by HuSkin reference

Table 20: Source-level falsification analysis comparing matched-source and different-source calibration.

x
/Users/chayada_p/Data/CUDENT/202604_skin_permeability_qspr_model/R/tables/Table_S10_wrong_source_by_refere

11.11 Supplementary Figure S1. Applicability-domain diagnostics

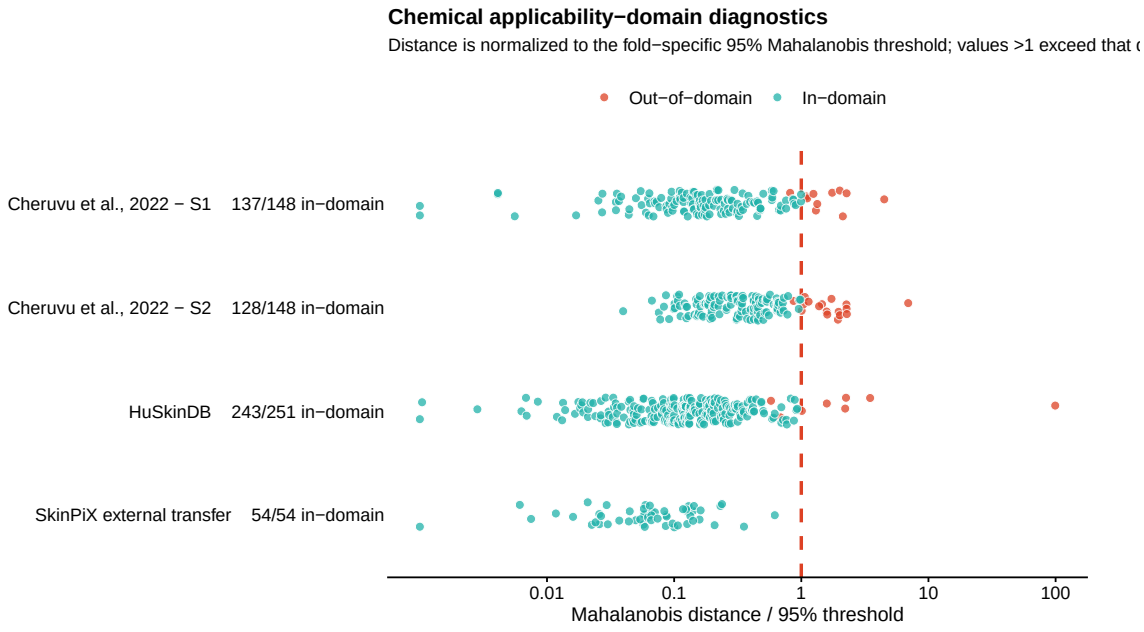


Figure 5: Applicability-domain diagnostics across Cheruvu S1, Cheruvu S2, HuSkinDB, and the external SkinPiX transfer set. Mahalanobis distance is normalized to the relevant 95% threshold; classification also requires all descriptors to remain within their training-set univariate ranges.

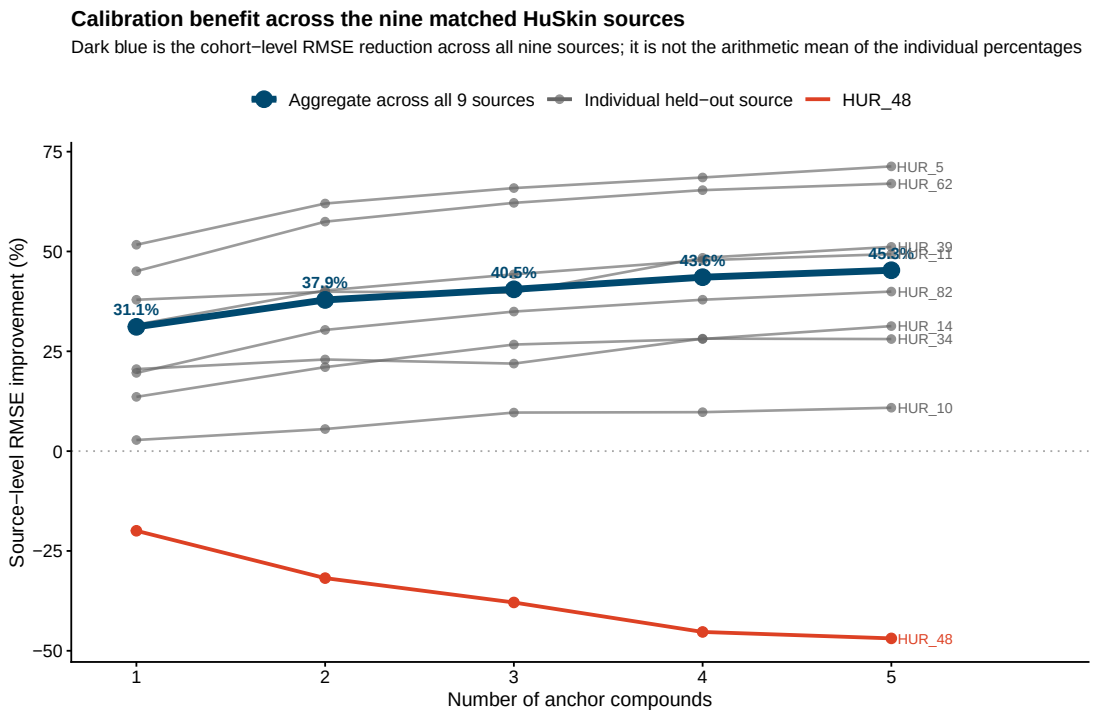


Figure 6: Calibration benefit across the nine matched HuSkin sources. Grey trajectories show individual held-out sources, the dark-blue trajectory is the cohort-level reduction in mean source-level RMSE calculated across all nine sources, and HUR_48 is highlighted as the persistent calibration failure. The aggregate trajectory is not the arithmetic mean of the source-level percentage improvements.

11.12 Supplementary Figure S2. HuSkin source-level calibration heterogeneity

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